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SUPERSEDING
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18 August 1976

MILITARY STANDARD

COMPRESSORS, HERMETIC: FOR USE WITH REFRIGERANT 12

FSC 4130

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DEPARTMENT OF DEFENSE

WASHINGTON, DC 20325

COMPRESSORS, HERMETIC: FOR USE WITH

REFRIGERANT 12

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2. Recommended corrections, additions, or deletions should be addressed to the US Army Natick Research and Development Command, Natick, MA 01760.

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CONTENTS

Paragraph		Page
1.	SCOPE	1
1.1	Scope	1
1.1.1	Application	1
1.2	Classification	1
2.	REFERENCED DOCUMENTS	1
2.1	Governmental	1
3.	DEFINITIONS	1
3.1	Hermetic compressor	1
4.	GENERAL REQUIREMENTS	2
4.1	Procurement specification	2
5.	DETAIL REQUIREMENTS	2
5.1	Refrigerant	2
5.2	Capacities	2
5.3	Operating conditions	2
5.4	Physical characteristics	3
5.5	Electrical characteristics	3

FIGURES

1.	Dimensions and weights	4
2.	Motor characteristics	5
3.	Motor controls and protectors	6
4.	Motor starting relays - mounting position and brackets	7
5.	Connection sizes	8
6.	Mounting plates	9

TABLES

I.	Compressor capacities in Btu/hr	2

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1. SCOPE

1.1 Scope. This standard covers capacities, rating conditions, and physical and electrical characteristics of standard hermetic compressors in a nominal horsepower range of 1/3 to 7-1/2, inclusive.

1.1.1 Application. The hermetic compressors covered by this standard are for use in combination with other standard refrigeration components in refrigeration and air-conditioning systems which employ refrigerant 12.

1.2 Classification. The hermetic compressors covered by this standard shall be classified in eight sizes according to capacities, as shown in table I.

2. REFERENCED DOCUMENTS

2.1 Governmental. The issues of the following documents in effect on the date of invitation for bids or request for proposal, form a part of this standard to the extent specified herein:

SPECIFICATION

MILITARY

MIL-C-23365 - Compressors, Hermetic: For Use With Refrigerant 12.

STANDARD

MILITARY

MS 17248 - Valves, Compressor Service, Refrigerant 12.

3. DEFINITIONS

3.1 Hermetic compressor. A unit in which the compressor and compressor motor are enclosed in the same housing with no external shaft or shaft seal, and the motor operates in a refrigerant atmosphere. The hermetic compressor housing may be the fully welded, the brazed, or the service-sealed type. In the fully welded or brazed type, the housing is permanently sealed and is not provided with a means of access for servicing in the field. In the service-sealed type, the housing is provided with some means of access for servicing internal parts in the field.

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4. GENERAL REQUIREMENTS

4.1 Procurement specification. The procurement specification for compressors covered by this standard is MIL-C-23365.

5. DETAIL REQUIREMENTS

5.1 Refrigerant. Compressors shall be for use with refrigerant 12 (dichlorodifluoromethane) (CCl_2F_2).

5.2 Capacities. Compressors shall have the capacities shown in table I.

TABLE I. Compressor capacities Btuh (Joule (J)) 1/

Size	Saturated refrigerant suction vapor temperatures		
	-10F (-23.5C)	+25F (-3.8C)	+50F (10C)
H-1	900 (949KJ) 2/	2,500 (2637KJ)	- - -
H-2	1,200 (1266KJ)	3,400 (3587KJ)	- - -
H-3	1,800 (1899KJ)	4,800 (5064KJ)	- - -
H-4	2,400 (2532KJ)	6,500 (6857KJ)	10,700 (11288KJ)
H-5	4,600 (4853KJ)	13,700 (14453KJ)	23,000 (24265KJ)
H-6	5,900 (6224KJ)	16,800 (17724KJ)	30,500 (32177KJ)
H-7	9,000 (9495KJ)	29,200 (30806KJ)	50,700 (53488KJ)
H-8	12,400 (13082KJ)	40,200 (42411KJ)	73,000 (77015KJ)

1/ Capacities may vary plus or minus 15 percent.

2/ 1055 Joules = 1 Btu.

5.3 Operating conditions. Compressors shall have the capacities shown in table I when operating in an ambient temperature of plus 110F (43.3C), at a condensing temperature of plus 135F (57.2C), corrected to 0F (17.7C) subcooling, and at applicable suction temperatures as follows:

Saturated refrigerant suction vapor temperature	Actual suction vapor temperature
-10°F (-23.5C)	+50°F (10C)
0°F (17.7C) and above	+65°F (18.3C)

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5.4 Physical characteristics. Overall compressor dimensions and maximum weights of compressors shall be as shown on figure 1. The size of suction and discharge connections shall be as shown on figure 5. Compressor mounting plates and location of suction and discharge connections shall be as shown on figure 6.

5.5 Electrical characteristics. Electrical characteristics of compressor motors shall be as shown on figure 2. Motor controls and protectors shall be as shown on figure 3. Motor relay mounting brackets shall be as shown on figure 4.

Notice of availability. Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.

Copies of this standard for military use may be obtained as indicated in the foreword to, or the general provisions of, the Index of Military Specifications and Standards.

The title and identifying symbol should be stipulated when requesting copies of military standards.

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COMPRESSOR ASSEMBLY				
SIZE	COMPRESSOR OVERALL DIMENSIONS (IN.) (MAX)			WEIGHT (LB) (MAX) (SEE NOTE 1)
	LENGTH	WIDTH	HEIGHT	
H-1	17	13	13	105
H-2	17	13	13	105
H-3	17	13	13	115
H-4	17	13	13	115
H-5	26 3/4	17	18	215
H-6	26 3/4	17	18	265
H-7	26 3/4	17	20	315
H-8	26 3/4	17	20	375

NOTE: 1. INCLUDES COMPRESSOR CHARGED WITH OIL, MOUNTING PLATES, CONTROLS, PIPING, FLANGES, AND SERVICE VALVES

FIGURE-1. DIMENSIONS AND WEIGHTS

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MOTORS

SIZE	ELECTRICAL TYPE	RATING		
		VOLTS	PHASE	CYCLES
H-1	CAPACITOR START	115	1	60
H-2	CAPACITOR START	115	1	60
H-3	CAPACITOR START	230	1	60
H-4	CAPACITOR START CAPACITOR RUN	230	1	60
H-5	INDUCTION	208/230	3	60
H-6	INDUCTION	208/230	3	60
H-7	INDUCTION	208/230	3	60
H-8	INDUCTION	208/230	3	60

Capacitors shall conform to the manufacturer's standard product provided the rest of the requirements for the hermetic compressors are met. Any changes made to capacitors, thereafter, must be resubmitted for approval.

FIGURE-2 MOTOR CHARACTERISTICS

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REQUIREMENTS AND LOCATIONS						
SIZE	CONTROLS TABLE I			MINIMUM MOTOR PROTECTION (1) TABLE II		TABLE III
	STARTING CAPACITOR	RUNNING CAPACITOR	CAPACITOR RELAYS	OVER CURRENT PROTECTOR	OVER TEMPERATURE PROTECTOR (3)	COMBINATION PROTECTOR (3) (4)
H-1	ON COMPRESSOR OR REMOTE (5)	NONE	ON COMPRESSOR OR REMOTE (5)	ON COMPRESSOR OR REMOTE (4)(5)	ON COMPRESSOR	ON COMPRESSOR
H-2						
H-3		ON COMPRESSOR OR REMOTE (5)				
H-4						
H-5 (2)	NONE	NONE	NONE	REMOTE (4)(5)		
H-6 (2)						
H-7 (2)						
H-8 (2)						

NOTES

1. HERMETIC COMPRESSORS WHICH DO NOT HAVE INHERENT MOTOR PROTECTION SHALL BE EQUIPPED WITH MOTOR PROTECTION SHOWN IN EITHER TABLE II OR TABLE III.

2. ALL PROTECTORS WHICH ARE INFLUENCED BY MOTOR CURRENT SHALL BE PLACED IN A MINIMUM OF TWO (2) PHASES OF ALL THREE (3) PHASE MOTORS.

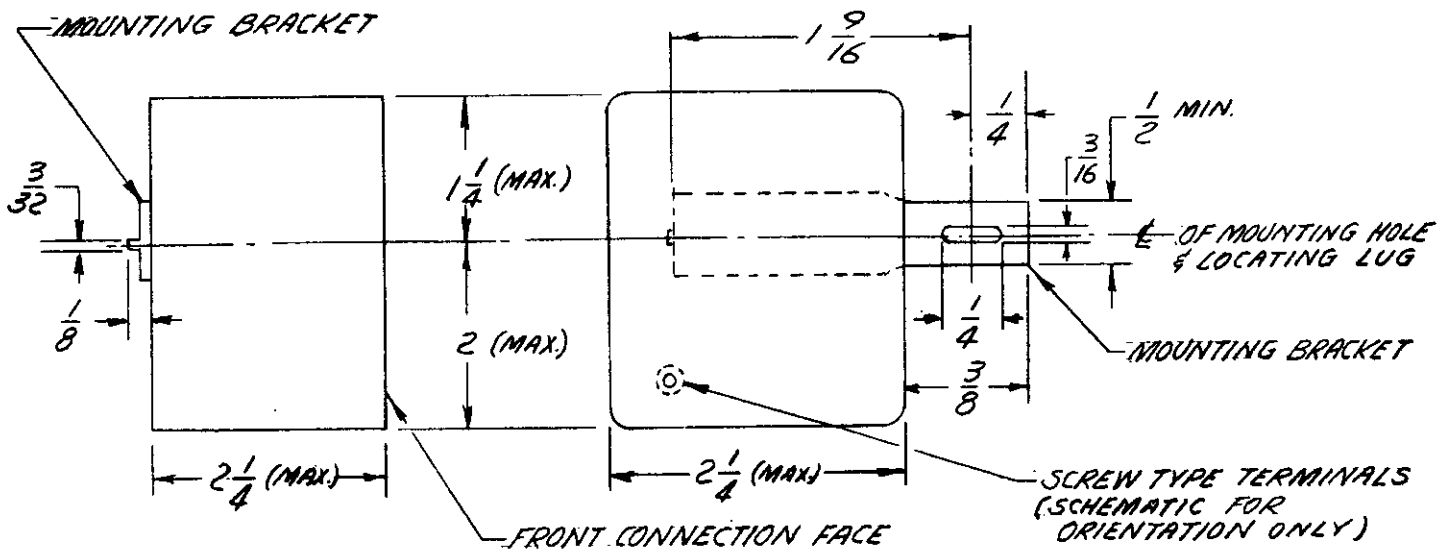
3. ALL PROTECTORS THAT ARE INFLUENCED BY MOTOR TEMPERATURE SHALL BE LOCATED ON THE COMPRESSORS OF THE FULLY WELDED TYPE; ON THE MOTOR HOUSING OF THE SERVICE SEALED TYPE; OR IN, ON, OR ADJACENT TO THE MOTOR WINDINGS AND IN SUCH A POSITION THAT THEY RECEIVE THE MAXIMUM EFFECT OF MOTOR TEMPERATURE.

4. COMPRESSORS WHICH REQUIRE MORE THAN ONE (1) SET OF HEATER COILS TO PROTECT THE MOTOR FROM CURRENT OVERLOADS WHEN OPERATING THRU THE MILITARY STANDARD SATURATED SUCTION TEMPERATURE RANGE SHALL BE EQUIPPED WITH A MOUNTED PROTECTIVE DEVICE (OVER CURRENT OR COMBINATION) FOR EACH HEATER COIL RATING NECESSARY. THESE PROTECTIVE DEVICES SHALL HAVE THEIR TERMINALS PLAINLY MARKED WITH THE SATURATED SUCTION TEMPERATURE RANGE FOR WHICH EACH PROTECTOR IS TO BE USED.

5. REMOTE CONTROLS AND PROTECTORS SHALL BE MOUNTED IN A SINGLE DRIP PROOF CONTROL BOX WHICH SHALL BE CONNECTED TO THE COMPRESSOR WITH FOUR (4) FT OF ARMORED CABLE.

FIGURE – 3 MOTOR CONTROLS AND PROTECTORS

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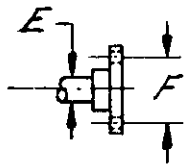


NOTES:

1. RELAY CASE SHALL BE SECURED TO MOUNTING BRACKET BY MANUFACTURER'S COMMERCIAL METHOD.
2. RELAY WILL BE MOUNTED ON A VERTICAL PLATE WITH SCREW TAB TO RIGHT FACING FRONT (CONNECTION) FACE. OTHER END OF BRACKET SHALL HAVE POSITTIONING LUG AS SHOWN.
3. MOUNTING BRACKET CONFIGURATION (BACK OF RELAY) SHOWN FOR ILLUSTRATIVE PURPOSES ONLY AND SHALL BE SAME AS MANUFACTURER'S COMMERCIAL BRACKET.
4. MATERIAL OF MOUNTING BRACKETS SHALL BE NONCORROSIVE AND SAME GAUGE AS MANUFACTURER'S COMMERCIAL PRODUCT.

FIGURE 4. MOTOR STARTING RELAYS - MOUNTING POSITION AND BRACKETS

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VIEW SHOWING
FLANGE CONNECTIONS

FLANGE DIMENSIONS		
E TUBE SIZE (OD)	F BOLT HOLE ℄ DIM	DIA BOLT HOLE
3/8, 1/2, 5/8, 7/8	1 5/8	11/32 ± 1/64
1 1/8	2 3/4	17/32 ± 1/64

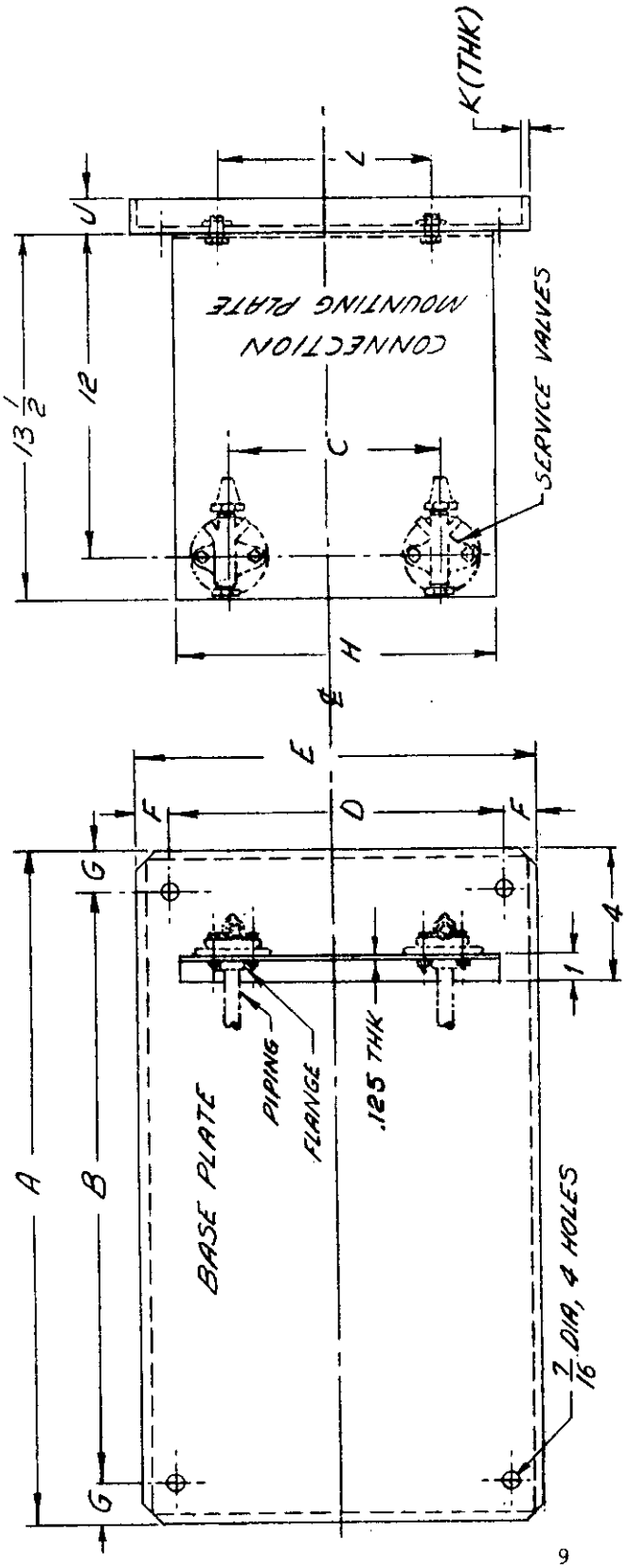
	H-1	H-2	H-3	H-4	H-5	H-6	H-7	H-8
SIZE (O.D.) DISCHARGE CONNECTION	3/8	3/8	3/8	1/2	5/8	5/8	7/8	7/8
SIZE (O.D.) SUCTION CONNECTION	3/8	1/2	1/2	5/8	7/8	1 1/8	1 1/8	1 1/8

NOTES:

1. ADEQUATE TUBING OF PROPER TYPE AND SIZE SHALL BE FURNISHED WITH EACH COMPRESSOR TO ABSORB COMPRESSOR VIBRATION. COMPRESSORS REQUIRING TUBING OF 5/8 IN. O.D. OR LARGER SHALL HAVE VIBRATION ELIMINATORS INSTALLED.
2. DIMENSIONS ARE IN INCHES.

FIGURE 5. CONNECTION SIZES

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COMPRESSOR SIZE	A	B	C	D	E	F	G	H	J	K	L
H-1, H-2, H-3, H-4	21 1/2	19 1/4	7	10 1/4	13	1 3/8	1 1/8	10	1	.190	7
H-5, H-6	31	26 1/2	8	13	17	2	2 1/4	14	1 3/8	250	8
H-7, H-8	31	26 1/2	10	13	17	2	2 1/4	14	1 3/8	250	

NOTES:
1. MATERIAL - ALUMINUM
2. SERVICE VALVES IN ACCORDANCE WITH MS17248
3. ALL DIMENSIONS IN INCHES

FIGURE - 6 MOUNTING PLATES

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